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AI-Driven Semantic Mapping and Explainable EHR Integration for AYUSH and ICD-11

Abstract

Healthcare systems across the world rely on standardized medical terminologies to enable accurate diagnosis, interoperability, and data exchange. In India, the AYUSH healthcare system uses traditional medical terminologies that are not directly compatible with global classification systems such as the International Classification of Diseases (ICD-11). This creates challenges in integrating AYUSH clinical data into modern Electronic Health Record (EHR) and Electronic Medical Record (EMR) systems.

This project presents an AI-driven approach for semantic mapping between AYUSH disease concepts and ICD-11 (Traditional Medicine Module-2) using explainable artificial intelligence. The system uses structured datasets, machine learning-assisted semantic reasoning, and a human-in-the-loop review mechanism to ensure accuracy and clinical safety. The solution is designed as an EHR-ready system with explainable outputs and approval workflows, making it suitable for both academic research and real-world healthcare applications.

Introduction

Electronic Health Records play a crucial role in modern healthcare by enabling standardized data storage, clinical decision support, and interoperability across systems. While ICD-11 is widely adopted internationally, AYUSH systems in India follow distinct terminologies that are semantically rich but not standardized for global interoperability.

Manual mapping between AYUSH terms and ICD-11 codes is time-consuming and prone to inconsistencies. With the increasing availability of artificial intelligence techniques, especially large language models, it is now possible to automate semantic mapping while maintaining transparency and human oversight.

This project explores the use of AI-assisted semantic mapping combined with explainable outputs and clinical validation to bridge the gap between AYUSH terminologies and ICD-11, supporting digital health initiatives and EHR integration.

Objectives

The main objectives of this project are to develop an automated semantic mapping system between AYUSH and ICD-11 terminologies, to ensure explainability of AI-generated mappings, to incorporate human validation for clinical safety, and to design an architecture that can be integrated into EHR and EMR systems. The project also aims to demonstrate how such a system can be extended to real-world healthcare datasets in production environments.

Data Description

For the purpose of this academic project, **sample datasets** representing AYUSH disease terms and ICD-11 disease concepts were used. These datasets were structured in CSV format and included disease names and brief clinical descriptions. The use of sample data allows safe experimentation, repeatability, and academic evaluation.

However, the system architecture and implementation are fully capable of handling **real-world healthcare data**. In a production environment, the same pipeline can be connected to authenticated government portals, WHO ICD-11 APIs, or licensed clinical datasets without any change in system logic. This ensures that the project is not limited to sample data and can be scaled for real EHR deployment.

icd11_terms.csv:

icd11_code	icd11_term	description
RA01	Rheumatoid arthritis	Chronic autoimmune inflammatory joint disease
OA01	Osteoarthritis	Degenerative joint disease
DB01	Diabetes mellitus	Metabolic disorder characterized by hyperglycemia
GS01	Gastritis	Inflammation of the gastric mucosa
IBS01	Irritable bowel syndrome	Functional gastrointestinal disorder
FEV01	Fever	Elevated body temperature due to illness
AST01	Asthma	Chronic inflammatory airway disease
AN01	Anemia	Reduced oxygen-carrying capacity of blood
HEM01	Hemorrhoids	Swollen veins in the anal region
DI01	Diarrhea	Increased frequency of loose stools
UR01	Urinary retention	Inability to empty bladder
UR02	Painful urination	Dysuria due to infection or inflammation
SK01	Psoriasis	Chronic inflammatory skin disease
SK02	Eczema	Inflammatory skin condition causing itching
BL01	Bleeding disorder	Abnormal bleeding tendency
ED01	Edema	Abnormal fluid accumulation in tissues
TB01	Tuberculosis	Chronic infectious lung disease
MH01	Psychotic disorder	Severe mental disorder with loss of reality
NE01	Epilepsy	Chronic neurological seizure disorder
NE02	Sciatica	Nerve pain radiating along sciatic nerve
PA01	Hemiplegia	Paralysis of one side of body
GI01	Dyspepsia	Indigestion and upper abdominal discomfort
GI02	Loss of appetite	Reduced desire to eat
GI03	Vomiting	Forceful expulsion of stomach contents

icd11_terms

ayush_terms.csv:

ayush_term	description
Amavata	Inflammatory joint disorder caused by accumulation of ama and vata imbalance
Sandhivata	Degenerative joint disorder characterized by pain and stiffness
Prameha	Metabolic disorder involving excessive urination and glucose imbalance
Amlapitta	Gastric disorder caused by excess pitta leading to acidity
Grahani	Chronic digestive disorder affecting intestinal absorption
Jwara	Febrile condition caused by infection or dosha imbalance
Kasa	Respiratory disorder characterized by persistent cough
Tamaka Shwasa	Chronic respiratory disorder comparable to bronchial asthma
Pandu	Anemia-like condition characterized by pallor and fatigue
Arsha	Anorectal disorder similar to hemorrhoids
Atisara	Acute or chronic diarrheal disorder
Udavarta	Disorder caused by upward movement of vata
Hridroga	Cardiac disorder affecting heart function
Mutrakrichra	Urinary disorder with painful micturition
Mutraghata	Urinary retention disorder
Kustha	Chronic skin disorder including psoriasis and eczema
Visarpa	Spreading inflammatory skin disease
Raktapitta	Bleeding disorder caused by vitiated blood
Shotha	Inflammatory swelling or edema
Kshaya	Wasting disease comparable to tuberculosis
Rajayakshma	Chronic infectious wasting disease
Unmada	Mental disorder affecting cognition and behavior
Apasmara	Neurological disorder similar to epilepsy
Gridhrasi	Neurological pain disorder comparable to sciatica

ayush_terms

semantic_mapping.csv:

ayush_term	ayush_description	icd11_term	confidenc	reason
Amavata	Inflammatory joint disorder caused by accumulation of ama and vata imba	Bone wastin	0.1	Amavata is described as an 'inflammatory joint disorder'. The provided ICD-11 options
Jwara	Febrile condition caused by infection or dosha imbalance	Mixed infect	0.95	Jwara is described as a 'febrile condition caused by infection.' The ICD-11 term 'Mixed i
Kasa	Respiratory disorder characterized by persistent cough	Asthma	0.9	Kasa is described as a 'Respiratory disorder characterized by persistent cough.' Asthma
Tamaka Shwasa	Chronic respiratory disorder comparable to bronchial asthma		0	The AYUSH disease 'Tamaka Shwasa' is described as a 'Chronic respiratory disorder con
Udavarta	Disorder caused by upward movement of vata	Dyspepsia	0.9	Udavarta in Ayurveda refers to the upward or abnormal movement of Vata, often due t
Mutrakrichra	Urinary disorder with painful micturition	Urinary rete	0.7	The AYUSH disease 'Mutrakrichra' is described as a 'Urinary disorder with painful mictu
Mutraghata	Urinary retention disorder	Urinary rete	1	The AYUSH description 'Urinary retention disorder' directly matches the ICD-11 term 'U
Kustha	Chronic skin disorder including psoriasis and eczema	Taste disord	0	Kustha is described as a 'Chronic skin disorder including psoriasis and eczema'. None of
Chardi	Vomiting disorder	Pediatric dia	0.2	Chardi is described as a 'Vomiting disorder', which is a gastrointestinal symptom-based
Trishna	Excessive thirst disorder	Urinary trac	0.7	Trishna describes an 'Excessive thirst disorder'. While 'excessive thirst' (polydipsia) is a
Aruchi	Loss of appetite disorder	Loss of appe	1	The AYUSH disease 'Aruchi' is described as 'Loss of appetite disorder'. The ICD-11 term
Hikka	Hiccup disorder	Gastritis	0.1	The AYUSH term 'Hikka' describes a hiccup disorder. No direct semantic equivalent for '

Methodology

The project follows a modular and layered methodology. Initially, AYUSH and ICD-11 datasets are prepared in structured formats. These datasets are then processed using an AI-based semantic mapping module powered by a large language model. The AI model analyzes disease names and descriptions to identify the most appropriate ICD-11 concept for each AYUSH term.

Each AI-generated mapping includes a confidence score and a natural-language explanation describing the reasoning behind the mapping. Based on the confidence score, mappings are either auto-approved, flagged for human review, or rejected. This confidence-based logic ensures controlled automation.

A human-in-the-loop mechanism is implemented using a review interface where clinicians or reviewers can approve, reject, or override AI suggestions. All decisions are recorded with reviewer details and timestamps, ensuring traceability and auditability.

Explainable AI features are incorporated to provide deeper insights, counterfactual explanations, alternative mappings, and bias or uncertainty analysis. These features enhance trust and transparency in AI-assisted clinical systems.

System Architecture

The system follows a clear and scalable architecture. Structured datasets serve as the input layer. The AI semantic mapping engine processes these inputs and generates explainable outputs. A lightweight database stores mappings and review metadata. A FastAPI backend exposes EHR-ready APIs, while a Streamlit-based interface enables interactive review and explainability. The system also supports exporting approved mappings in standard formats suitable for healthcare interoperability.

Technologies Used

The project is implemented using Python as the primary programming language. FastAPI is used to develop the backend API services, while SQLite serves as the database for storing mappings and review information. Streamlit is used to build the interactive explainable AI dashboard. Pandas is used for data handling and preprocessing. The Gemini API is used to perform semantic reasoning and generate explainable outputs. The system design aligns with healthcare standards such as ICD-11, Traditional Medicine Module-2, and FHIR-based interoperability principles.

Explainable AI and Human-in-the-Loop Design

Explainability is a key aspect of this project. The AI model provides natural-language explanations for each mapping, allowing users to understand why a particular ICD-11 concept was selected. Additional explainability features include deeper explanations on demand, counterfactual reasoning that highlights conditions under which a mapping may fail, and alternative ICD suggestions with reasoning.

Human oversight is enforced throughout the system. AI outputs do not directly modify clinical records. Instead, reviewers validate mappings before they are approved for use in EHR systems. This design follows best practices in clinical decision support systems and ensures ethical and responsible use of AI.

Results and Observations

The system successfully demonstrated semantic mapping between AYUSH and ICD-11 concepts using sample datasets. The explainable AI interface allowed reviewers to understand and validate AI decisions effectively. The confidence-based review workflow reduced manual effort while maintaining accuracy and safety. The architecture proved flexible, scalable, and suitable for real-world EHR integration.

Advantages of the System

The proposed system reduces manual effort in terminology mapping, improves consistency, and enhances transparency through explainable AI. It supports human validation, making it safe for clinical environments. The modular design allows easy integration with existing EHR and EMR systems and can be extended to other medical terminologies.

Limitations

The accuracy of semantic mapping depends on the quality of input descriptions. As the project uses sample data, real-world performance may vary based on data richness and completeness. The AI model operates as a black-box system and does not expose internal weights, limiting model-level interpretability.

Future Scope

Future work includes integrating real-world authenticated healthcare datasets, implementing multi-language support for AYUSH terminologies, adding patient-level EHR workflows, incorporating authentication mechanisms such as ABHA, and deploying the system in cloud environments. Further enhancements may also include performance benchmarking and integration with national health information systems.

Conclusion

This Biopython course project demonstrates how AI-assisted semantic mapping can bridge traditional and modern medical terminologies in a safe, explainable, and scalable manner. By combining semantic reasoning, human oversight, and EHR-ready design, the project provides a practical solution for healthcare interoperability challenges. Although sample data was used for implementation, the system is fully capable of handling real-world datasets, making it relevant for both academic and production-level healthcare applications.

Screenshots and Output

Screenshots of the Streamlit interface, mapping outputs, and explainable AI features are included at the end of this report.

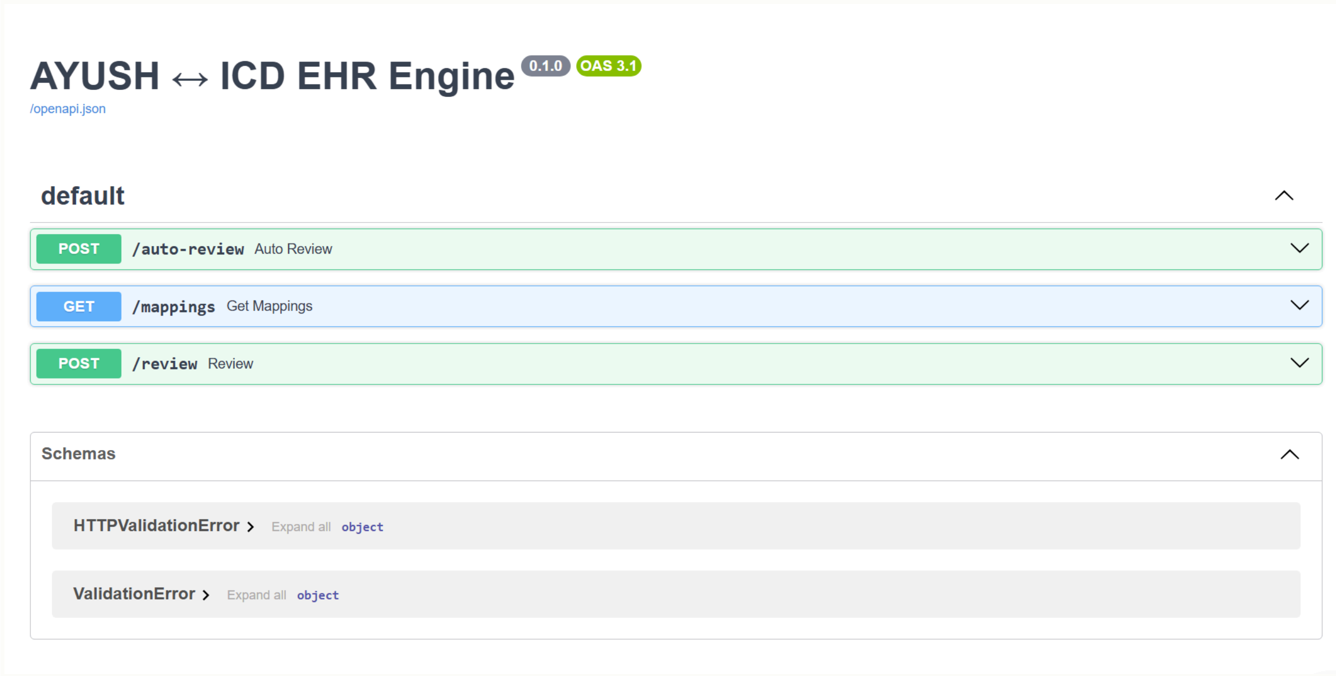
(emrehr) C:\Users\saniy>python C:\Users\saniy\OneDrive\Desktop\BioPython_CourseProject\database.py

✅ Database initialized

📁 Location: ehr.db

(emrehr) C:\Users\saniy>python C:\Users\saniy\OneDrive\Desktop\BioPython_CourseProject\load_mapping.py

✅ Real mappings loaded into DB


AYUSH ↔ ICD EHR Engine 0.1.0 OAS 3.1
/openapi.json
default
POST /auto-review Auto Review
GET /mappings Get Mappings
POST /review Review
Schemas
HTTPValidationError Expand all object
ValidationError Expand all object


AYUSH ↔ ICD EHR Engine 0.1.0 OAS 3.1
/openapi.json
default
POST /auto-review Auto Review
Parameters
Try it out
No parameters
Responses
Code
Description
Links
200
Successful Response
No links
Media type
application/json
Controls Accept header.
Example Value
Schema
"string"

GET `/mappings` Get Mappings

Parameters

Cancel

Name	Description
status string *(query)*	PENDING

Execute

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

POST /review Review

Parameters

Try it out

Name	Description
id * required integer *(query)*	id
action * required string *(query)*	action
reviewer * required string *(query)*	reviewer

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	*No links*

Media type

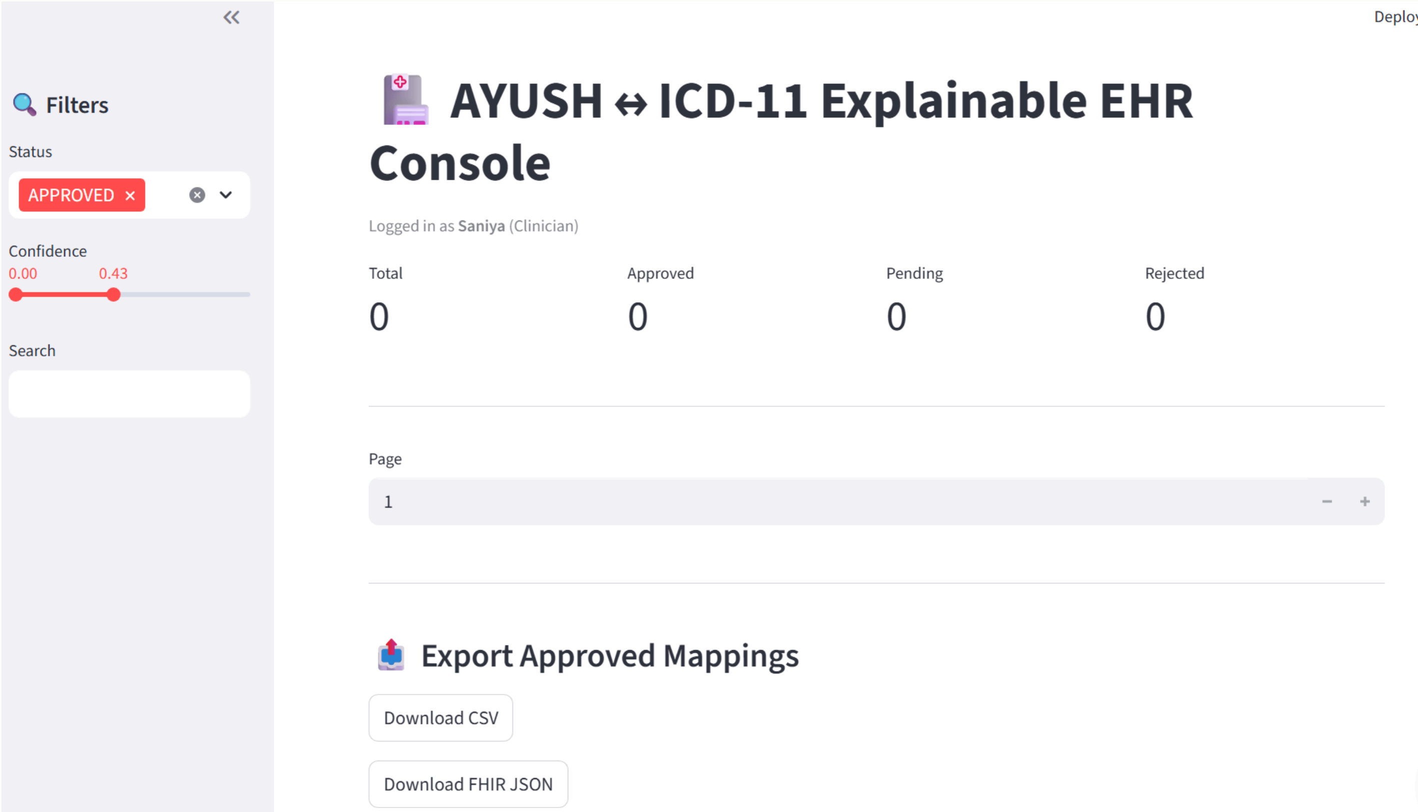
application/json

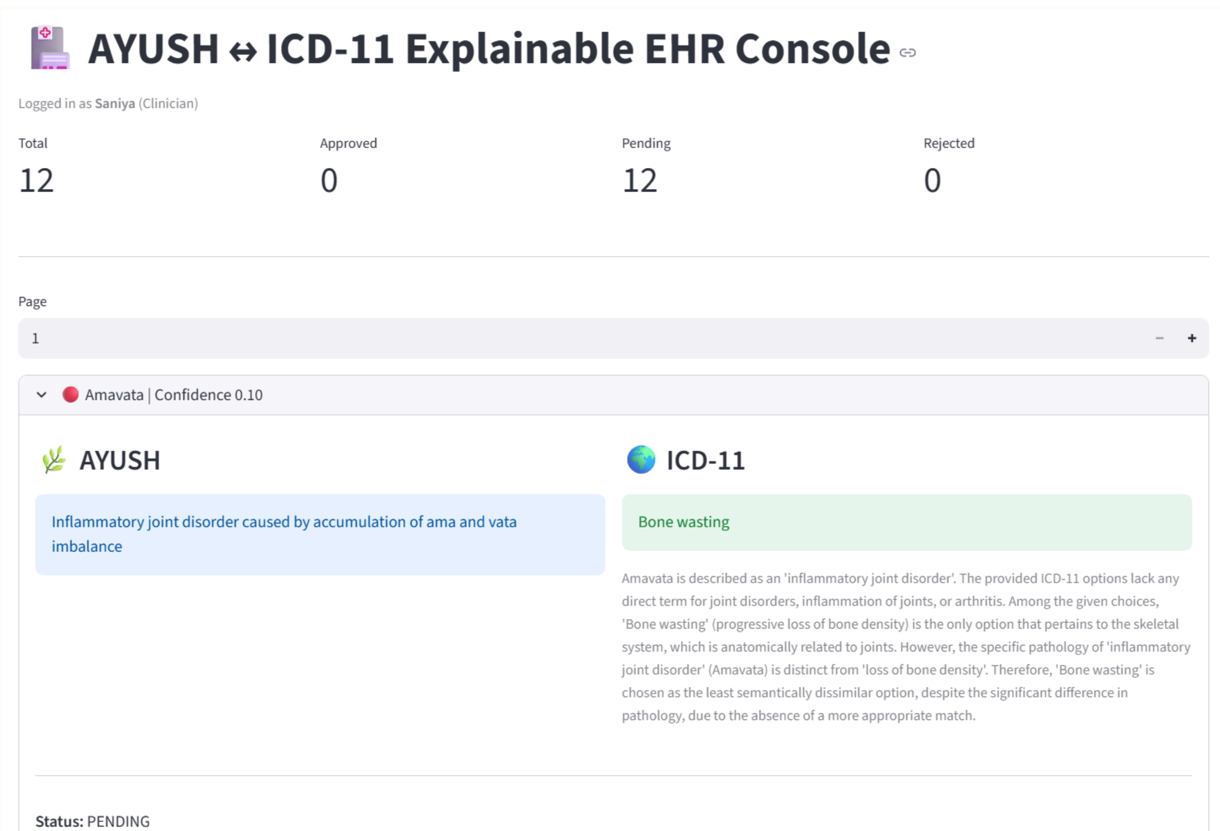
Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

Schemas

HTTPValidationError > Expand all object


Deploy
Filters
Status
APPROVED
Confidence
0.00
0.43
Search
AYUSH ↔ ICD-11 Explainable EHR Console
Logged in as Saniya (Clinician)
Total
0
Approved
0
Pending
0
Rejected
0
Page
1
Export Approved Mappings
Download CSV
Download FHIR JSON


AYUSH ↔ ICD-11 Explainable EHR Console
Logged in as Saniya (Clinician)
Total
12
Approved
0
Pending
12
Rejected
0
Page
1
Amavata | Confidence 0.10
AYUSH
Inflammatory joint disorder caused by accumulation of ama and vata imbalance
ICD-11
Bone wasting
Amavata is described as an 'inflammatory joint disorder'. The provided ICD-11 options lack any direct term for joint disorders, inflammation of joints, or arthritis. Among the given choices, 'Bone wasting' (progressive loss of bone density) is the only option that pertains to the skeletal system, which is anatomically related to joints. However, the specific pathology of 'inflammatory joint disorder' (Amavata) is distinct from 'loss of bone density'. Therefore, 'Bone wasting' is chosen as the least semantically dissimilar option, despite the significant difference in pathology, due to the absence of a more appropriate match.
Status: PENDING

Total	Approved	Pending	Rejected
12	0	12	0

Page

1

- Amavata | Confidence 0.10
- Jwara | Confidence 0.95
- Kasa | Confidence 0.90
- Tamaka Shwasa | Confidence 0.00
- Udavarta | Confidence 0.90
- Mutrakrichra | Confidence 0.70

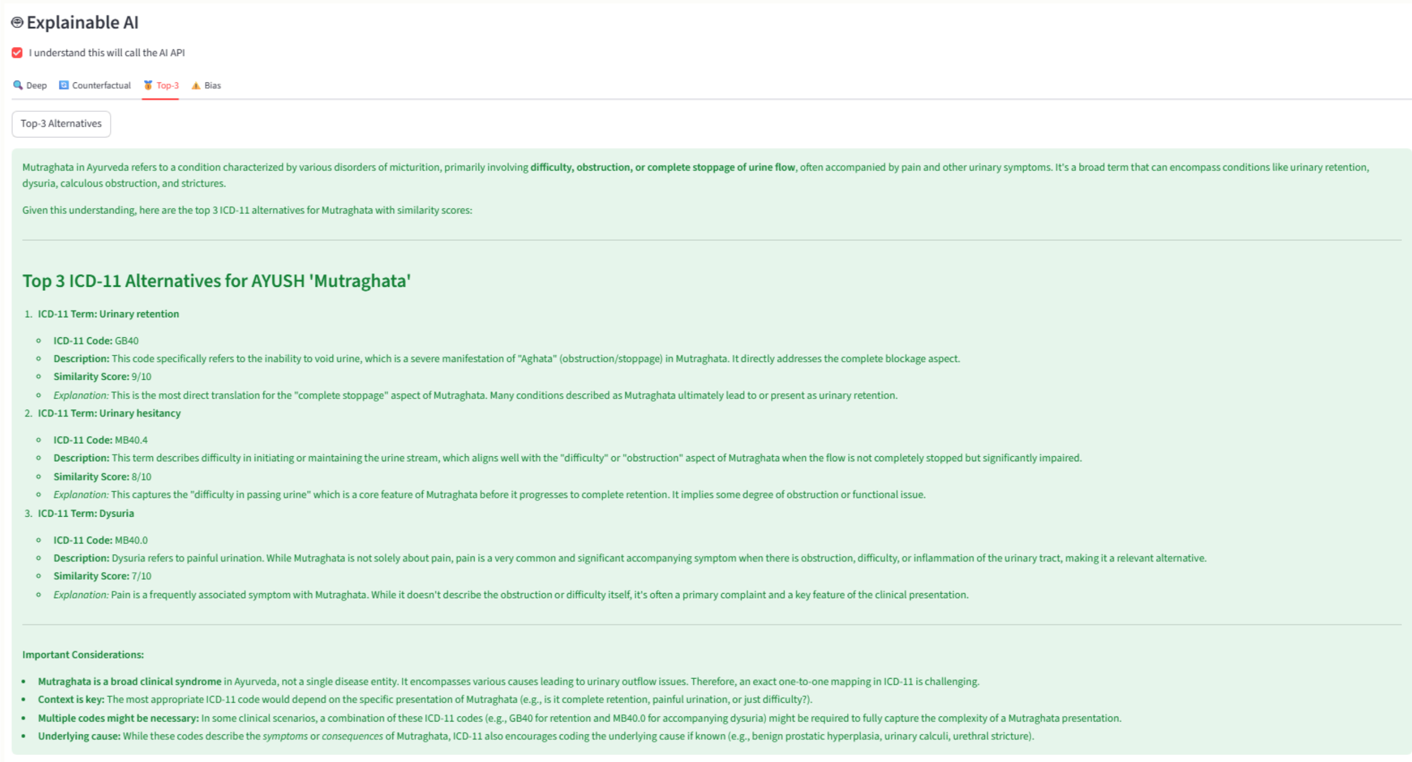
📝 Clinical Decision

Override ICD-11 Term

Urinary retention

✅ Approve

❌ Reject



Explainable AI

I understand this will call the AI API

Deep | Counterfactual | Top-3 | Bias

Top-3 Alternatives

Mutraghata in Ayurveda refers to a condition characterized by various disorders of micturition, primarily involving **difficulty, obstruction, or complete stoppage of urine flow**, often accompanied by pain and other urinary symptoms. It's a broad term that can encompass conditions like urinary retention, dysuria, calculous obstruction, and strictures.

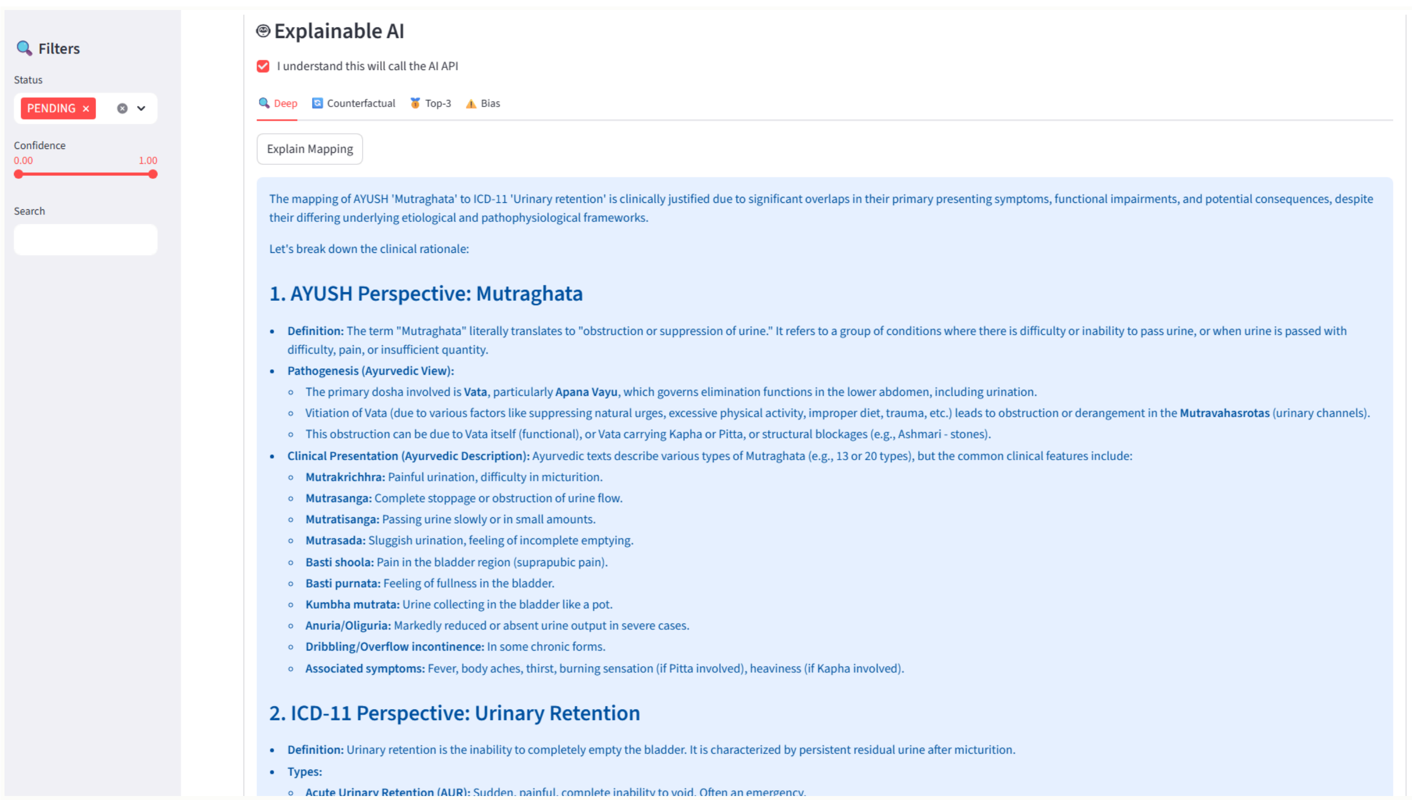
Given this understanding, here are the top 3 ICD-11 alternatives for Mutraghata with similarity scores:

Top 3 ICD-11 Alternatives for AYUSH 'Mutraghata'

1. **ICD-11 Term: Urinary retention**
    - **ICD-11 Code:** GB40
    - **Description:** This code specifically refers to the inability to void urine, which is a severe manifestation of "Aghata" (obstruction/stoppage) in Mutraghata. It directly addresses the complete blockage aspect.
    - **Similarity Score:** 9/10
    - *Explanation:* This is the most direct translation for the "complete stoppage" aspect of Mutraghata. Many conditions described as Mutraghata ultimately lead to or present as urinary retention.
2. **ICD-11 Term: Urinary hesitancy**
    - **ICD-11 Code:** MB40.4
    - **Description:** This term describes difficulty in initiating or maintaining the urine stream, which aligns well with the "difficulty" or "obstruction" aspect of Mutraghata when the flow is not completely stopped but significantly impaired.
    - **Similarity Score:** 8/10
    - *Explanation:* This captures the "difficulty in passing urine" which is a core feature of Mutraghata before it progresses to complete retention. It implies some degree of obstruction or functional issue.
3. **ICD-11 Term: Dysuria**
    - **ICD-11 Code:** MB40.0
    - **Description:** Dysuria refers to painful urination. While Mutraghata is not solely about pain, pain is a very common and significant accompanying symptom when there is obstruction, difficulty, or inflammation of the urinary tract, making it a relevant alternative.
    - **Similarity Score:** 7/10
    - *Explanation:* Pain is a frequently associated symptom with Mutraghata. While it doesn't describe the obstruction or difficulty itself, it's often a primary complaint and a key feature of the clinical presentation.

Important Considerations:

- **Mutraghata is a broad clinical syndrome** in Ayurveda, not a single disease entity. It encompasses various causes leading to urinary outflow issues. Therefore, an exact one-to-one mapping in ICD-11 is challenging.
- **Context is key:** The most appropriate ICD-11 code would depend on the specific presentation of Mutraghata (e.g., is it complete retention, painful urination, or just difficulty?).
- **Multiple codes might be necessary:** In some clinical scenarios, a combination of these ICD-11 codes (e.g., GB40 for retention and MB40.0 for accompanying dysuria) might be required to fully capture the complexity of a Mutraghata presentation.
- **Underlying cause:** While these codes describe the *symptoms* or *consequences* of Mutraghata, ICD-11 also encourages coding the underlying cause if known (e.g., benign prostatic hyperplasia, urinary calculi, urethral stricture).



Filters

Status

PENDING ×

Confidence

0.00 1.00

Search

Explainable AI

I understand this will call the AI API

Deep | Counterfactual | Top-3 | Bias

Explain Mapping

The mapping of AYUSH 'Mutraghata' to ICD-11 'Urinary retention' is clinically justified due to significant overlaps in their primary presenting symptoms, functional impairments, and potential consequences, despite their differing underlying etiological and pathophysiological frameworks.

Let's break down the clinical rationale:

1. AYUSH Perspective: Mutraghata

- **Definition:** The term "Mutraghata" literally translates to "obstruction or suppression of urine." It refers to a group of conditions where there is difficulty or inability to pass urine, or when urine is passed with difficulty, pain, or insufficient quantity.
- **Pathogenesis (Ayurvedic View):**
    - The primary dosha involved is **Vata**, particularly **Apana Vayu**, which governs elimination functions in the lower abdomen, including urination.
    - Vitiation of Vata (due to various factors like suppressing natural urges, excessive physical activity, improper diet, trauma, etc.) leads to obstruction or derangement in the **Mutravahasrotas** (urinary channels).
    - This obstruction can be due to Vata itself (functional), or Vata carrying Kapha or Pitta, or structural blockages (e.g., Ashmari - stones).
- **Clinical Presentation (Ayurvedic Description):** Ayurvedic texts describe various types of Mutraghata (e.g., 13 or 20 types), but the common clinical features include:
    - **Mutrakrichhra:** Painful urination, difficulty in micturition.
    - **Mutrasanga:** Complete stoppage or obstruction of urine flow.
    - **Mutratisanga:** Passing urine slowly or in small amounts.
    - **Mutrasada:** Sluggish urination, feeling of incomplete emptying.
    - **Basti shoola:** Pain in the bladder region (suprapubic pain).
    - **Basti purnata:** Feeling of fullness in the bladder.
    - **Kumbha mutrata:** Urine collecting in the bladder like a pot.
    - **Anuria/Oliguria:** Markedly reduced or absent urine output in severe cases.
    - **Dribbling/Overflow incontinence:** In some chronic forms.
    - **Associated symptoms:** Fever, body aches, thirst, burning sensation (if Pitta involved), heaviness (if Kapha involved).

2. ICD-11 Perspective: Urinary Retention

- **Definition:** Urinary retention is the inability to completely empty the bladder. It is characterized by persistent residual urine after micturition.
- **Types:**
    - **Acute Urinary Retention (AUR):** Sudden, painful, complete inability to void. Often an emergency.

AYUSH rows loaded: 91

ICD rows loaded: 82

0%| | 0/91 [00:00<?, ?it/s]✔ Amavata → Bone wasting

1%|█ | 1/91 [00:09<14:21, 9.58s/it]❌ FAILED Sandhivata: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

2%|██ | 2/91 [00:22<17:04, 11.51s/it]❌ FAILED Prameha: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

3%|███ | 3/91 [00:26<12:02, 8.21s/it]❌ FAILED Amlapitta: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

4%|████ | 4/91 [00:29<08:45, 6.04s/it]❌ FAILED Grahani: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

5%|█████ | 5/91 [00:31<06:51, 4.78s/it]✔ Jwara → Mixed infection fever

7%|██████ | 6/91 [00:37<07:07, 5.03s/it]✔ Kasa → Asthma

8%|███████ | 7/91 [00:59<14:43, 10.51s/it]✔ Tamaka Shwasa →

9%|████████ | 8/91 [01:08<14:09, 10.23s/it]❌ FAILED Pandu: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

10%|█████████ | 9/91 [01:22<15:31, 11.35s/it]❌ FAILED Arsha: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/usage?tab=rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 41.390686914s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric':

'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '41s'}]}}

11%|██████████ | 10/91 [01:23<11:06, 8.23s/it]❌
FAILED Atisara: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/usage?tab=rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 40.166165059s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '40s'}]}}

12%|███████████ | 11/91 [01:25<08:06, 6.09s/it]✔
Udavarta → Dyspepsia

13%|████████████ | 12/91 [01:42<12:20, 9.38s/it]❌
FAILED Hridroga: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'The model is overloaded. Please try again later.', 'status': 'UNAVAILABLE'}}

14%|█████████████ | 13/91 [01:44<09:27, 7.28s/it]✔
Mutrakrichra → Urinary retention

15%|██████████████ | 14/91 [01:52<09:41, 7.55s/it]✔
Mutraghata → Urinary retention

16%|███████████████ | 15/91 [01:57<08:29, 6.70s/it]✔
Kustha → Taste disorder

18%|████████████████ | 16/91 [02:09<10:28, 8.38s/it]